

DBMS LAB

JOIN SQL COMMANDS

CREATING STUDENT AND DEPARTMENT TABLES:

[SQL Worksheet]*      Aa 

```
1 CREATE TABLE Student (
2     student_id INT PRIMARY KEY,
3     student_name VARCHAR(50),
4     dept_id INT
5 );
6 CREATE TABLE Department (
7     dept_id INT PRIMARY KEY,
8     dept_name VARCHAR(50)
9 );
```

Query result Script output DBMS output Explain Plan SQL history

SQL> CREATE TABLE Student (
 student_id INT PRIMARY KEY,
 student_name VARCHAR(50),
 dept_id INT...
[Show more...](#)

Table STUDENT created.

Elapsed: 00:00:00.010

[SQL Worksheet]*      Aa 

```
6 CREATE TABLE Department (
7     dept_id INT PRIMARY KEY,
8     dept_name VARCHAR(50)
9 );
10 INSERT INTO Student (student_id, student_name, dept_id)
11 VALUES
12 (101, 'Rahul', 10),
13 (102, 'Priya', 20),
14 (103, 'Arjun', 10),
```

Query result Script output DBMS output Explain Plan SQL history

SQL> CREATE TABLE Department (
 dept_id INT PRIMARY KEY,
 dept_name VARCHAR(50)
)

Table DEPARTMENT created.

Elapsed: 00:00:00.010

INSERTING VALUES INTO STUDENT AND DEPARTMENT TABLES :

```
10 INSERT INTO Student (student_id, student_name, dept_id)
11 VALUES
12 (101, 'Rahul', 10),
13 (102, 'Priya', 20),
14 (103, 'Arjun', 10),
15 (104, 'Sneha', 30),
16 (105, 'Kiran', 40);
17 INSERT INTO Department (dept_id, dept_name)
```

Query result Script output DBMS output Explain Plan SQL



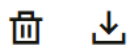
```
SQL> INSERT INTO Student (student_id, student_name, dept_id)
VALUES
(101, 'Rahul', 10),
(102, 'Priya', 20),...
Show more...
```

5 rows inserted.

Elapsed: 00:00:00.142

```
17 INSERT INTO Department (dept_id, dept_name)
18 VALUES
19 (10, 'CSE'),
20 (20, 'ECE'),
21 (30, 'EEE'),
22 (50, 'MECH');
23 select student_id, student_name, dept_name from student, DEPARTMENT;
24
```

Query result Script output DBMS output Explain Plan SQL history



```
SQL> INSERT INTO Department (dept_id, dept_name)
VALUES
(10, 'CSE'),
(20, 'ECE'),...
Show more...
```

4 rows inserted.

Elapsed: 00:00:00.058

CARTIESAN JOIN :

```
22 (50, 'MECH');
23 select student_id, student_name, dept_name from student, DEPARTMENT;
24
25 select student.student_id,
```

Query result Script output DBMS output Explain Plan SQL history

  Download Execution time: 0.002 seconds

	STUDENT_ID	STUDENT_NAME	DEPT_NAME
8	103	Arjun	ECE
9	104	Sneha	ECE
10	105	Kiran	ECE
11	101	Rahul	EEE
12	102	Priya	EEE
13	103	Arjun	EEE

EQUI JOIN :

```
25 select student.student_id,
26 student.student_name,
27 department.dept_id,
28 department.dept_name
29 FROM student, department
30 WHERE student.dept_id = department.dept_id;
31
32 select * from student;
```

Query result Script output DBMS output Explain Plan SQL history

  Download Execution time: 0.001 seconds

	STUDENT_ID	STUDENT_NAME	DEPT_ID	DEPT_NAME
1	101	Rahul	10	CSE
2	102	Priya	20	ECE
3	103	Arjun	10	CSE
4	104	Sneha	30	EEE

SELF JOIN:

```
32 select * from student;
33 SELECT s1.student_name AS student1,
34        s2.student_name AS student2,
35        s1.dept_id
36 FROM student s1, student s2
37 WHERE s1.dept_id = s2.dept_id;
38 select student_id, student_name, dept_id from student;
```

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.008 seconds

	STUDENT1	STUDENT2	DEPT_ID
2	Arjun	Rahul	10
3	Priya	Priya	20
4	Rahul	Arjun	10
5	Arjun	Arjun	10
6	Sneha	Sneha	30

OUTER JOIN:

```
38 select student_id, student_name, dept_id from student;
39 SELECT a.student_id,
40        a.student_name,
41        b.dept_id,
42        b.dept_name
43 FROM student a, department b
44 WHERE a.dept_id(+) = b.dept_id;
45
```

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.01 seconds

	STUDENT_ID	STUDENT_NAME	DEPT_ID	DEPT_NAME
1	101	Rahul	10	CSE
2	102	Priya	20	ECE
3	103	Arjun	10	CSE
4	104	Sneha	30	EEE
5	(null)	(null)	50	MECH